

```
/mvnw text eol=lf
*.cmd text eol=crlf
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license agreements. See the NOTICE file# distributed with this work for additional
information# regarding copyright ownership. The ASF licenses this file# to you under
the Apache License, Version 2.0 (the# "License"); you may not use this file except in
compliance# with the License. You may obtain a copy of the License at## http://
www.apache.org/licenses/LICENSE-2.0## Unless required by applicable law or agreed
to in writing,# software distributed under the License is distributed on an# "AS IS"
BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY# KIND, either express or
implied. See the License for the# specific language governing permissions and
limitations# under the License.wrapperVersion3.3.2distributionTypeonly-script
distributionUrlhttps://repo.maven.apache.org/maven2/org/apache/maven/apache-
maven/3.9.9/apache-maven-3.9.9-bin.zip
```

```
=
=
=
```

```
#!/bin/sh
# -----
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# regarding copyright ownership. The ASF licenses this file
# to you under the Apache License, Version 2.0 (the
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```

```

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# KIND, either express or implied. See the License for the
# specific language governing permissions and limitations
# under the License.
# -----

# -----
# Apache Maven Wrapper startup batch script, version 3.3.2
#
# Optional ENV vars
# -----
# JAVA_HOME - location of a JDK home dir, required when download maven via java
source
# MVNW_REPOURL - repo url base for downloading maven distribution
# MVNW_USERNAME/MVNW_PASSWORD - user and password for downloading
maven
# MVNW_VERBOSE - true: enable verbose log; debug: trace the mvnw script; others:
silence the output
# -----

set -euf
[ "${MVNW_VERBOSE-}" != debug ] || set -x

# OS specific support.
native_path() { printf %s\n "$1"; }
case "$(uname)" in
CYGWIN* | MINGW*)
    [ -z "${JAVA_HOME-}" ] || JAVA_HOME="$(cygpath --unix "$JAVA_HOME")"
    native_path() { cygpath --path --windows "$1"; }
    ;;
esac

# set JAVACMD and JAVACCMD
set_java_home() {
    # For Cygwin and MinGW, ensure paths are in Unix format before anything is touched
    if [ -n "${JAVA_HOME-}" ]; then
        if [ -x "$JAVA_HOME/jre/sh/java" ]; then
            # IBM's JDK on AIX uses strange locations for the executables
            JAVACMD="$JAVA_HOME/jre/sh/java"
            JAVACCMD="$JAVA_HOME/jre/sh/javac"
        else
            JAVACMD="$JAVA_HOME/bin/java"
            JAVACCMD="$JAVA_HOME/bin/javac"
        fi
    else
        JAVACMD="$JAVA_HOME/bin/java"
        JAVACCMD="$JAVA_HOME/bin/javac"
    fi

    if [ ! -x "$JAVACMD" ] || [ ! -x "$JAVACCMD" ]; then
        echo "The JAVA_HOME environment variable is not defined correctly, so mvnw

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cannot run." >&2
    echo "JAVA_HOME is set to \"${JAVA_HOME}\", but \"${JAVA_HOME}/bin/java\" or \"${
JAVA_HOME}/bin/javac\" does not exist." >&2
    return 1
fi
fi
else
    JAVACMD="$(
    'set' +e
    'unset' -f command 2>/dev/null
    'command' -v java
    )" || :
    JAVACCMD="$(
    'set' +e
    'unset' -f command 2>/dev/null
    'command' -v javac
    )" || :

    if [ ! -x "${JAVACMD}" ] || [ ! -x "${JAVACCMD}" ]; then
        echo "The java/javac command does not exist in PATH nor is JAVA_HOME set, so
mvnw cannot run." >&2
        return 1
    fi
fi
}

# hash string like Java String::hashCode
hash_string() {
    str="${1:-}" h=0
    while [ -n "$str" ]; do
        char="${str%${str#?}}"
        h=$((h * 31 + $(LC_CTYPE=C printf %d "$char"))) % 4294967296
        str="${str#?}"
    done
    printf %x\\n $h
}

verbose() { ::; }
[ "${MVNW_VERBOSE}" != true ] || verbose() { printf %s\\n "${1-}"; }

die() {
    printf %s\\n "$1" >&2
    exit 1
}

trim() {

```

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# MWRAPPER-139:
# Trims trailing and leading whitespace, carriage returns, tabs, and linefeeds.
# Needed for removing poorly interpreted newline sequences when running in more
# exotic environments such as mingw bash on Windows.
printf "%s" "${1}" | tr -d '[:space:]'
}

# parse distributionUrl and optional distributionSha256Sum, requires .mvn/wrapper/
maven-wrapper.properties
while IFS="=" read -r key value; do
    case "${key-}" in
        distributionUrl) distributionUrl=$(trim "${value-}") ;;
        distributionSha256Sum) distributionSha256Sum=$(trim "${value-}") ;;
    esac
done <"${0%/*}/.mvn/wrapper/maven-wrapper.properties"
[ -n "${distributionUrl-}" ] || die "cannot read distributionUrl property in ${0%/*}/.mvn/
wrapper/maven-wrapper.properties"

case "${distributionUrl##*/}" in
maven-mvnd-*.bin.*)
    MVN_CMD=mvnd.sh _MVNW_REPO_PATTERN=/maven/mvnd/
    case "${PROCESSOR_ARCHITECTURE-}${PROCESSOR_ARCHITEW6432-}:
$(uname -a)" in
        *AMD64:CYGWIN* | *AMD64:MINGW*) distributionPlatform=windows-amd64 ;;
        :Darwin*x86_64) distributionPlatform=darwin-amd64 ;;
        :Darwin*arm64) distributionPlatform=darwin-aarch64 ;;
        :Linux*x86_64*) distributionPlatform=linux-amd64 ;;
        *)
            echo "Cannot detect native platform for mvnd on $(uname)-$(uname -m), use pure
java version" >&2
            distributionPlatform=linux-amd64
            ;;
    esac
    distributionUrl="${distributionUrl%-bin.*}-${distributionPlatform}.zip"
    ;;
maven-mvnd-*) MVN_CMD=mvnd.sh _MVNW_REPO_PATTERN=/maven/mvnd/ ;;
*) MVN_CMD="mvn${0##*/mvnw}" _MVNW_REPO_PATTERN=/org/apache/maven/ ;;
esac

# apply MVNW_REPOURL and calculate MAVEN_HOME
# maven home pattern: ~/.m2/wrapper/dists/{apache-maven-<version>,maven-mvnd-
<version>-<platform>}/<hash>
[ -z "${MVNW_REPOURL-}" ] || distributionUrl="$MVNW_REPOURL$_MVNW_REPO_
PATTERN${distributionUrl##*_MVNW_REPO_PATTERN}"
distributionUrlName="${distributionUrl##*/}"
distributionUrlNameMain="${distributionUrlName%.*}"

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distributionUrlNameMain="${distributionUrlNameMain%-bin}"
MAVEN_USER_HOME="${MAVEN_USER_HOME:-${HOME}/.m2}"
MAVEN_HOME="${MAVEN_USER_HOME}/wrapper/dists/${distributionUrlNameMain-}/
$(hash_string "$distributionUrl")"

exec_maven() {
    unset MVNW_VERBOSE MVNW_USERNAME MVNW_PASSWORD
    MVNW_REPOURL || :
    exec "$MAVEN_HOME/bin/$MVN_CMD" "$@" || die "cannot exec $MAVEN_HOME/
bin/$MVN_CMD"
}

if [ -d "$MAVEN_HOME" ]; then
    verbose "found existing MAVEN_HOME at $MAVEN_HOME"
    exec_maven "$@"
fi

case "${distributionUrl}" in
    *?-bin.zip | *?maven-mvnd-*?*.zip) ;;
    *) die "distributionUrl is not valid, must match *-bin.zip or maven-mvnd-*.zip, but found
'${distributionUrl}'" ;;
esac

# prepare tmp dir
if TMP_DOWNLOAD_DIR="$(mktemp -d)" && [ -d "$TMP_DOWNLOAD_DIR" ]; then
    clean() { rm -rf -- "$TMP_DOWNLOAD_DIR"; }
    trap clean HUP INT TERM EXIT
else
    die "cannot create temp dir"
fi

mkdir -p -- "${MAVEN_HOME%/*}"

# Download and Install Apache Maven
verbose "Couldn't find MAVEN_HOME, downloading and installing it ..."
verbose "Downloading from: $distributionUrl"
verbose "Downloading to: $TMP_DOWNLOAD_DIR/$distributionUrlName"

# select .zip or .tar.gz
if ! command -v unzip >/dev/null; then
    distributionUrl="${distributionUrl%.zip}.tar.gz"
    distributionUrlName="${distributionUrl##*/}"
fi

# verbose opt
__MVNW_QUIET_WGET=---quiet __MVNW_QUIET_CURL=---silent

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__MVNW_QUIET_UNZIP=-q __MVNW_QUIET_TAR="
[ "${MVNW_VERBOSE-}" != true ] || __MVNW_QUIET_WGET="
__MVNW_QUIET_CURL=" __MVNW_QUIET_UNZIP=" __MVNW_QUIET_TAR=v

# normalize http auth
case "${MVNW_PASSWORD:+has-password}" in
  ") MVNW_USERNAME=" MVNW_PASSWORD=" ;;
  has-password) [ -n "${MVNW_USERNAME-}" ] || MVNW_USERNAME="
  MVNW_PASSWORD=" ;;
  esac

if [ -z "${MVNW_USERNAME-}" ] && command -v wget >/dev/null; then
  verbose "Found wget ... using wget"
  wget ${__MVNW_QUIET_WGET:+"${__MVNW_QUIET_WGET"}"} "$distributionUrl" -O
"$TMP_DOWNLOAD_DIR/$distributionUrlName" || die "wget: Failed to fetch
$distributionUrl"
elif [ -z "${MVNW_USERNAME-}" ] && command -v curl >/dev/null; then
  verbose "Found curl ... using curl"
  curl ${__MVNW_QUIET_CURL:+"${__MVNW_QUIET_CURL"}"} -f -L -o
"$TMP_DOWNLOAD_DIR/$distributionUrlName" "$distributionUrl" || die "curl: Failed to
fetch $distributionUrl"
elif set_java_home; then
  verbose "Falling back to use Java to download"
  javaSource="$TMP_DOWNLOAD_DIR/Downloader.java"
  targetZip="$TMP_DOWNLOAD_DIR/$distributionUrlName"
  cat >"$javaSource" <<-END
  public class Downloader extends java.net.Authenticator
  {
    protected java.net.PasswordAuthentication getPasswordAuthentication()
    {
      return new
java.net.PasswordAuthentication( System.getenv( "MVNW_USERNAME" ),
System.getenv( "MVNW_PASSWORD" ).toCharArray() );
    }
    public static void main( String[] args ) throws Exception
    {
      setDefault( new Downloader() );
      java.nio.file.Files.copy( java.net.URI.create( args[0] ).toURL().openStream(),
java.nio.file.Paths.get( args[1] ).toAbsolutePath().normalize() );
    }
  }
  END
  # For Cygwin/MinGW, switch paths to Windows format before running javac and java
  verbose " - Compiling Downloader.java ..."
  "$(native_path "$JAVACCMD")" "$(native_path "$javaSource")" || die "Failed to
compile Downloader.java"

```

```

verbose " - Running Downloader.java ..."
"${native_path "$JAVACMD")" -cp "${native_path "$TMP_DOWNLOAD_DIR")"
Downloader "$distributionUrl" "${native_path "$targetZip")"
fi

# If specified, validate the SHA-256 sum of the Maven distribution zip file
if [ -n "${distributionSha256Sum-}" ]; then
    distributionSha256Result=false
    if [ "$MVN_CMD" = mvnd.sh ]; then
        echo "Checksum validation is not supported for maven-mvnd." >&2
        echo "Please disable validation by removing 'distributionSha256Sum' from your
maven-wrapper.properties." >&2
        exit 1
    elif command -v sha256sum >/dev/null; then
        if echo "$distributionSha256Sum $TMP_DOWNLOAD_DIR/$distributionUrlName" |
sha256sum -c >/dev/null 2>&1; then
            distributionSha256Result=true
        fi
    elif command -v shasum >/dev/null; then
        if echo "$distributionSha256Sum $TMP_DOWNLOAD_DIR/$distributionUrlName" |
shasum -a 256 -c >/dev/null 2>&1; then
            distributionSha256Result=true
        fi
    else
        echo "Checksum validation was requested but neither 'sha256sum' or 'shasum' are
available." >&2
        echo "Please install either command, or disable validation by removing
'distributionSha256Sum' from your maven-wrapper.properties." >&2
        exit 1
    fi
    if [ $distributionSha256Result = false ]; then
        echo "Error: Failed to validate Maven distribution SHA-256, your Maven distribution
might be compromised." >&2
        echo "If you updated your Maven version, you need to update the specified
distributionSha256Sum property." >&2
        exit 1
    fi
fi

# unzip and move
if command -v unzip >/dev/null; then
    unzip ${__MVNW_QUIET_UNZIP:+"$__MVNW_QUIET_UNZIP"}
"$TMP_DOWNLOAD_DIR/$distributionUrlName" -d "$TMP_DOWNLOAD_DIR" || die
"failed to unzip"
else
    tar xzf${__MVNW_QUIET_TAR:+"$__MVNW_QUIET_TAR"}

```



```

@ ""=="" ( __MVNW_ARG0_NAME__=%~nx0)
@ __MVNW_CMD__=
@ __MVNW_ERROR__=
@ __MVNW_PSMODULEP_SAVE=
@ PSModulePath=
@ /F "usebackq tokens=* delims==" ( powershell -nopprofile "& {$scriptDir='%~dp0';
$script="; icm -ScriptBlock ([Scriptblock]::Create((Get-Content -Raw '%~f0'))) -
NoNewScope}") @(
    ""=="MVN_CMD" ( __MVNW_CMD__=) ""=="" ( ) ( =)
)
@ PSModulePath=
@ __MVNW_PSMODULEP_SAVE=
@ __MVNW_ARG0_NAME__=
@ MVNW_USERNAME=
@ MVNW_PASSWORD=
@ ""=="" ( %*)
@ Cannot maven from wrapper >& && /b
@ :EOF
: end batch / begin powershell #>

$ErrorActionPreference = "Stop"
($env:MVNW_VERBOSE -eq "true") {
    $VerbosePreference = "Continue"
}

# calculate distributionUrl, requires .mvn/wrapper/maven-wrapper.properties
$distributionUrl = (Get-Content -Raw "$scriptDir/.mvn/wrapper/maven-
wrapper.properties" | ConvertFrom-StringData).distributionUrl
(!$distributionUrl) {

```

```

    Write-Error "cannot read distributionUrl property $scriptDir/.mvn/wrapper/maven-
wrapper.properties"
}

```

```

switch -wildcard -casesensitive ( $($distributionUrl - '^.*/',") ) {
    "maven-mvnd-*" {
        $USE_MVND = $true
        $distributionUrl = $distributionUrl - '-bin\[^\.]*$', "-windows-amd64.zip"
        $MVN_CMD = "mvnd."
    }
    default {
        $USE_MVND = $false
        $MVN_CMD = $script - '^mvnw', 'mvn'
    }
}

```

```

# apply MVNW_REPOURL and calculate MAVEN_HOME
# maven home pattern: ~/.m2/wrapper/dists/{apache-maven-<version>,maven-mvnd-
<version>-<platform>}/{<hash>
($env:MVNW_REPOURL) {
    $MVNW_REPO_PATTERN = ($USE_MVND) { "/org/apache/maven/" } { "/maven/
mvnd/" }
    $distributionUrl =
"$env:MVNW_REPOURL$MVNW_REPO_PATTERN$($distributionUrl - '^.*'+
$MVNW_REPO_PATTERN,)"
}
$distributionUrlName = $distributionUrl - '^.*/',"
$distributionUrlNameMain = $distributionUrlName - '\[^\.]*$', " - '-bin$',"
$MAVEN_HOME_PARENT = "$HOME/.m2/wrapper/dists/$distributionUrlNameMain"
($env:MAVEN_USER_HOME) {
    $MAVEN_HOME_PARENT = "$env:MAVEN_USER_HOME/wrapper/dists/
$distributionUrlNameMain"
}
$MAVEN_HOME_NAME =
([System.Security.Cryptography.MD5]::Create().ComputeHash([byte[]
[char[]]$distributionUrl] | ForEach-Object {$_} -join "x2")) -join "
$MAVEN_HOME = "$MAVEN_HOME_PARENT/$MAVEN_HOME_NAME"

(Test- - "$MAVEN_HOME" -PathType Container) {
    Write-Verbose "found existing MAVEN_HOME $MAVEN_HOME"
    Write-Output "MVN_CMD=$MAVEN_HOME/bin/$MVN_CMD"
    $?
}

```

```
(! $distributionUrlNameMain -or ($distributionUrlName -eq $distributionUrlNameMain)) {
    Write-Error "distributionUrl is valid, must end with *-bin.zip, but found $distributionUrl"
}
```

```
# prepare tmp
$TMP_DOWNLOAD_DIR_HOLDER = New-TemporaryFile
$TMP_DOWNLOAD_DIR = New-Item -ItemType Directory -
"$TMP_DOWNLOAD_DIR_HOLDER."
$TMP_DOWNLOAD_DIR_HOLDER.Delete() | Out-Null
trap {
    ($TMP_DOWNLOAD_DIR.Exists) {
        try { Remove-Item $TMP_DOWNLOAD_DIR -Recurse -Force | Out-Null }
        catch { Write-Warning "Cannot remove $TMP_DOWNLOAD_DIR" }
    }
}
```

```
New-Item -ItemType Directory - "$MAVEN_HOME_PARENT" -Force | Out-Null
```

```
# Download and Install Apache Maven
Write-Verbose "Couldn't MAVEN_HOME, downloading and installing it ..."
Write-Verbose "Downloading from: $distributionUrl"
Write-Verbose "Downloading to: $TMP_DOWNLOAD_DIR/$distributionUrlName"
```

```
$webclient = New-Object System.WebClient
($env:MVNW_USERNAME -and $env:MVNW_PASSWORD) {
    $webclient.Credentials = New-Object
    System.Net.NetworkCredential($env:MVNW_USERNAME, $env:MVNW_PASSWORD)
}
[.ServicePointManager]::SecurityProtocol = [.SecurityProtocolType]::Tls12
$webclient.DownloadFile($distributionUrl, "$TMP_DOWNLOAD_DIR/
$distributionUrlName") | Out-Null
```

```
# specified, validate the SHA- sum of the Maven distribution zip file
$distributionSha256Sum = (Get-Content -Raw "$scriptDir/.mvn/wrapper/maven-
wrapper.properties" | ConvertFrom-StringData).distributionSha256Sum
($distributionSha256Sum) {
    ($USE_MVND) {
        Write-Error "Checksum validation is supported maven-mvnd. `nPlease disable
validation by removing 'distributionSha256Sum' from your maven-wrapper.properties."
    }
    Import-Module $PSHOME\Modules\Microsoft.PowerShell.Utility -Function Get-
FileHash
    ((Get-FileHash "$TMP_DOWNLOAD_DIR/$distributionUrlName" -Algorithm
SHA256).Hash.ToLower() -ne $distributionSha256Sum) {
        Write-Error "Error: Failed to validate Maven distribution SHA-, your Maven distribution
might be compromised. you updated your Maven version, you need to update the
```

specified distributionSha256Sum property."

```
}  
}
```

unzip and

```
Expand-Archive "$TMP_DOWNLOAD_DIR/$distributionUrlName" -DestinationPath  
"$TMP_DOWNLOAD_DIR" | Out-Null  
-Item - "$TMP_DOWNLOAD_DIR/$distributionUrlNameMain" -NewName  
$MAVEN_HOME_NAME | Out-Null  
try {  
  -Item - "$TMP_DOWNLOAD_DIR/$MAVEN_HOME_NAME" -Destination  
  $MAVEN_HOME_PARENT | Out-Null  
} catch {  
  (! (Test- - "$MAVEN_HOME" -PathType Container)) {  
    Write-Error "fail to MAVEN_HOME"  
  }  
} finally {  
  try { Remove-Item $TMP_DOWNLOAD_DIR -Recurse -Force | Out-Null }  
  catch { Write-Warning "Cannot remove $TMP_DOWNLOAD_DIR" }  
}
```

Write-Output "MVN_CMD=\$MAVEN_HOME/bin/\$MVN_CMD"

```
<?xml version="1.0"UTF-8"><projectxmlns="http://maven.apache.org/POM/4.0.0"  
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"xsi:schemaLocation"http://  
maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">  
  <modelVersion></modelVersion><parent><groupId></groupId><artifactId></artifactId><version></  
version><relativePath<!-- lookup parent from repository --></parent><groupId></groupId><  
artifactId></artifactId><version></version><name></name><description></description><url><  
licenses><license></licenses><developers><developer></developers><scm><connection><  
developerConnection><tag><url></scm><properties><java.version></java.version></properties><  
dependencies><dependency><groupId></groupId><artifactId></artifactId></dependency><  
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dependency><groupId></groupId><artifactId></artifactId></dependency><dependency><  
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=></span>  
></span>4.0.0></span>  
></span>  
></span>org.springframework.boot></span>
```

```

    ></span>spring-boot-starter-parent></span>
    ></span>3.4.1></span>
  /></span>
></span>
></span>com.codewithmosh></span>
></span>store></span>
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  ></span>spring-boot-starter-data-jpa></span>
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  ></span>mysql-connector-j></span>

```

```

        ></span>com.mysql></span>
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        ></span>flyway-core></span>
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        ></span>annotationProcessor></span>
    ></span>

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            ></span>org.flywaydb></span>
            ></span>flyway-maven-plugin></span>
            ></span>10.15.0></span>
            ></span>
                ></span>jdbc:mysql://localhost:3306/store?
createDatabaseIfNotExist=true></span>
                ></span>root></span>
                ></span>MyPassword!></span>
                ></span>>false></span>
            ></span>
        ></span>

        ></span>
            ></span>org.springframework.boot></span>
            ></span>spring-boot-maven-plugin></span>
        ></span>
    ></span>
></span>

# The Ultimate Spring Boot Course ``sh
git clone https://github.com/mosh-hamedani/spring-store

```

```
cd spring-store
````sh
git checkout part1-section3-start
``
```

This repository contains the source code for my Spring Boot course:

<https://codewithmosh.com/p/spring-boot-fundamentals>

Each commit in this repository is named after the corresponding lesson, so you can easily track changes and see exactly what code was written in each step of the course. This allows you to follow along lesson by lesson and compare your code with mine.

Additionally, I've added tags at the beginning of each major section. This makes it easy to jump to a specific section and start coding from there.

To get started, clone the repository to your local machine:

I've added tags at the beginning of each section. If you want to start from a particular section, you can check out the corresponding tag.

```
create table bigint primary key varchar255 not null varchar255 not null varchar255 not null
create table bigint primary key varchar255 not null varchar255 not null varchar255 not null
bigint not null constraint foreign key references users
(
 id auto_increment
 ,
 name () ,
 email () ,
 password ()
);

addresses
(
 id auto_increment
 ,
 street () ,
 city () ,
 zipcode () ,
 user_id ,
 addresses_users_id_fk
 (user_id) users (id)
);
```

```

package public interface ProductSummary {
 getId()
 getName()
}
com.codewithmosh.store.dtos;

{
 Long ;
 String ;
}
package import import @Getter @AllArgsConstructor public class ProductSummaryDTO
private private com.codewithmosh.store.dtos;

lombok.AllArgsConstructor;
lombok.Getter;

{
 Long id;
 String name;
}
package public interface UserSummary {
 getId()
 getEmail()
}
com.codewithmosh.store.dtos;

{
 Long ;
 String ;
}
package import import @Builder @AllArgsConstructor @NoArgsConstructor @ToString
@Getter @Setter @Entity @Table(name = "addresses") public class Address {
 @Id
 @GeneratedValue(strategy = GenerationType.IDENTITY) @Column(name = "id") private
 @Column(name = "street") private @Column(name = "city") private @Column(name =
 "zip") private @Column(name = "state") private @ManyToOne(fetch = FetchType.LAZY)
 @JoinColumn(name = "user_id") @ToString private com.codewithmosh.store.entities;

jakarta.persistence.*;
lombok.*;

{

```



Long id;

String street;

String city;

String zip;

String state;

.Exclude

User user;

```
}packageimportimportimportimportimportimportimport@Getter@Setter@NoArgsConstructor
@Entity@Table(name = "categories")publicclassCategory@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)@Column(name = "id")private
@Column(name = "name")private@OneToMany(mappedBy = "category")privatenew
HashSetpublicCategory(String name)thispublicCategory(byte)this
com.codewithmosh.store.entities;
```

```
jakarta.persistence.*;
lombok.Getter;
lombok.NoArgsConstructor;
lombok.Setter;
```

```
java.util.HashSet;
java.util.Set;
```

{

Byte id;

```

String name;

Set<Product> products = <>();

{
 .name = name;
}

id) {
 .id = id;
}
}packageimportpublicinterfaceCategoryRepositoryextendsCrudRepository
com.codewithmosh.store.entities;

org.springframework.data.repository.CrudRepository;

<Category, Byte> {
}packageimportimportimport@Getter@Setter@Builder@AllArgsConstructor
@NoArgsConstructor@Entity@Table(name = "products")publicclassProduct@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)@Column(name = "id")private
@Column(name = "name")private@Column(name = "description")private
@Column(name = "price")private@ManyToOne(cascade = CascadeType.PERSIST)
@JoinColumn(name = "category_id")private com.codewithmosh.store.entities;

jakarta.persistence.*;
lombok.*;

java.math.BigDecimal;

{

Long id;

```



<Product, Long>, ProductCriteriaRepository, JpaSpecificationExecutor<Product> {

List<Product> ;  
List<Product> ;  
List<Product> ;  
List<Product> ;  
List<Product> ;  
List<Product> ;  
List<Product> ;

List<Product> ;  
List<Product> ;  
List<Product> ;  
List<Product> ;  
List<Product> ;

List<Product> ;  
List<Product> ;

List<Product> ;

List<Product> ;

List<Product> ;  
List<Product> ;

List<Product> ;

BigDecimal min, BigDecimal max)</span>;

;

List<ProductSummaryDTO> Category category)</span>;

```

}packageimportimportimport@Builder@AllArgsConstructor@NoArgsConstructor
@ToString@Getter@Setter@Entity@Table(name = "profiles")publicclassProfile@Id
@Column(name = "id")private@Column(name = "bio")private@Column(name =
"phone_number")private@Column(name = "date_of_birth")private@Column(name =
"loyalty_points")private@OneToOne(fetch = FetchType.LAZY)@JoinColumn(name =
"id")@MapsIdprivate com.codewithmosh.store.entities;

```

```

jakarta.persistence.*;
lombok.*;

```

```

java.time.LocalDate;

```

```

{

```

```

 Long id;

```

```

 String bio;

```

```

 String phoneNumber;

```

```

 LocalDate dateOfBirth;

```

```

 Integer loyaltyPoints;

```

```

 User user;

```

```

}packageimportimportimportimportimportimportimportimport@Getter@Setter
@NoArgsConstructor@ToString@Entity@Table(name = "tags")publicclassTag@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)@Column(name = "id")private
@Column(name = "name")private@ManyToMany(mappedBy = "tags")@ToStringprivate
newHashSetpublicTag(String name)this com.codewithmosh.store.entities;

```



```

 name = "wishlist",
 joinColumns = @JoinColumn(name = "user_id"),
 inverseJoinColumns = @JoinColumn(name = "product_id")
)private new HashSet<>()public void addFavoriteProduct(Product product)@Override public
toString()return("id = ", "name = ", "email = ") com.codewithmosh.store.entities;

```

```

jakarta.persistence.*;
lombok.*;

```

```

java.util.ArrayList;
java.util.HashSet;
java.util.List;
java.util.Set;

```

```

{

```

```

 Long id;

```

```

 String name;

```

```

 String email;

```

```

 String password;

```

```

 .Default
 List<Address> addresses = <>();

```

```

 {
 addresses.add(address);
 address.setUser();
 }

```

```

 {

```

```

 addresses.remove(address);
 address.setUser();
 }

```

```

 {
 (tagName);
 tags.add(tag);
 tag.getUsers().add();
 }

```

```

.Default
Set<Tag> tags = <>();

```

Profile profile;

```

Set<Product> favoriteProducts = <>();

```

```

{
 favoriteProducts.add(product);
}

```

```

String {
 getClass().getSimpleName() + +
 + id + +
 + name + +
 + email + ;
}
}

```

```

packageimportimportpublicinterfaceAddressRepositoryextendsCrudRepository
com.codewithmosh.store.repositories;

```

```

com.codewithmosh.store.entities.Address;
org.springframework.data.repository.CrudRepository;

```

```

<Address, Long> {
}packageimportimportimportpublicinterfaceProductCriteriaRepository
findProductsByCriteria(String name, BigDecimal minPrice, BigDecimal maxPrice)
com.codewithmosh.store.repositories;

```

```

com.codewithmosh.store.entities.Product;

```



```

java.math.BigDecimal;
java.util.List;

{
 List<Product> ;
}
packageimportimportimportimportimportimportimportimportimportimportimportimport
@AllArgsConstructor@RepositorypublicclassProductCriteriaRepositoryImplimplements
ProductCriteriaRepository@PersistenceContextprivatefinal@Overridepublic
findProductsByCriteria(String name, BigDecimal minPrice, BigDecimal maxPrice)
CriteriaBuildercb=newArrayListifnull// name like %name%"name""%"%"ifnull// price >=
minPrice"price"ifnull// price <= maxPrice"price"newPredicatereturn
com.codewithmosh.store.repositories;

```

```

com.codewithmosh.store.entities.Product;
jakarta.persistence.EntityManager;
jakarta.persistence.PersistenceContext;
jakarta.persistence.criteria.CriteriaBuilder;
jakarta.persistence.criteria.CriteriaQuery;
jakarta.persistence.criteria.Predicate;
jakarta.persistence.criteria.Root;
lombok.AllArgsConstructor;
org.springframework.stereotype.Repository;

```

```

java.math.BigDecimal;
java.util.ArrayList;
java.util.List;

```

```

{

 EntityManager entityManager;

 List<Product> {
 entityManager.getCriteriaBuilder();
 CriteriaQuery<Product> cq = cb.createQuery(Product.class);
 Root<Product> root = cq.from(Product.class);

 List<Predicate> predicates = <>();
 (name !=) {

 predicates.add(cb.like(root.get(), + name +));
 }
 }
}

```

```
(minPrice !=) {

 predicates.add(cb.greaterThanOrEqualTo(root.get(), minPrice));
}
(maxPrice !=) {

 predicates.add(cb.lessThanOrEqualTo(root.get(), maxPrice));
}

cq.select(root).where(predicates.toArray([predicates.size()]));

entityManager.createQuery(cq).getResultList();
}
}
package import import import import import import import public interface ProfileRepository
extends CrudRepository com.codewithmosh.store.repositories;

com.codewithmosh.store.dtos.UserSummary;
com.codewithmosh.store.entities.Profile;
org.springframework.data.jpa.repository.EntityGraph;
org.springframework.data.jpa.repository.Query;
org.springframework.data.repository.CrudRepository;
org.springframework.data.repository.query.Param;

java.util.List;

<Profile, Long> {
} package import import import public class ProductSpec public static hasName(String name)
return "name""%" public static hasPriceGreaterThanOrEqualTo(BigDecimal price)
return "price" public static hasPriceLessThanOrEqualTo(BigDecimal price) return "price"
com.codewithmosh.store.repositories.specifications;

com.codewithmosh.store.entities.Product;
org.springframework.data.jpa.domain.Specification;

java.math.BigDecimal;

{
 Specification<Product> {
 (root, query, cb) -> cb.like(root.get(), + name +);
 }

 Specification<Product> {
 (root, query, cb) -> cb.greaterThanOrEqualTo(root.get(), price);
 }
}
```

```

Specification<Product> {
 (root, query, cb) -> cb.lessThanOrEqualTo(root.get(), price);
}
}

packageimportimportimportimportimportimportimportimportimportimportpublicinterface
UserRepositoryextendsCrudRepository@EntityGraph(attributePaths = {"tags",
"addresses"})findByEmail(String email)@EntityGraph(attributePaths = "addresses")
@Query("select u from User u")findAllWithTags()@Query("select u.id as id, u.email as
email from User u where u.profile.loyaltyPoints > :loyaltyPoints order by u.email")
findLoyalUsers(@Param("loyaltyPoints")int com.codewithmosh.store.repositories;

com.codewithmosh.store.dtos.UserSummary;
com.codewithmosh.store.entities.User;
org.springframework.data.jpa.repository.EntityGraph;
org.springframework.data.jpa.repository.Query;
org.springframework.data.repository.CrudRepository;
org.springframework.data.repository.query.Param;

java.util.List;
java.util.Optional;

<User, Long> {

Optional<User> ;

List<User> ;

List<UserSummary> loyaltyPoints);
}
packageimportimportimportimportimportimportimportimportimportimportimportimportimport
@AllArgsConstructor@ServicepublicclassUserServiceprivatefinalprivatefinalprivatefinal
privatefinalprivatefinalprivatefinal@TransactionalpublicvoidshowEntityStates()varuser=
"John Doe""john.doe@example.com""password"if"Persistent"else"Transient / Detached"
if"Persistent"else"Transient / Detached"@TransactionalpublicvoidshowRelatedEntities()
varprofile=2LpublicvoidfetchAddress()varaddress=1LpublicvoidpersistRelated()varuser=
"John Doe""john.doe@example.com""password"varaddress="street""city""state""zip"
@TransactionalpublicvoiddeleteRelated()varuser=3Lvaraddress=@Transactionalpublic
voidmanageProducts()4L@TransactionalpublicvoidupdateProductPrices()10byte1
@TransactionalpublicvoidfetchProducts()varproduct=newProduct"product"varmatcher=
"id""description"varexample=varproducts=publicvoidfetchProductsByCriteria()var
products="prod"1nullpublicvoidfetchProductsBySpecifications(String name, BigDecimal
minPrice, BigDecimal maxPrice)nullifnullifnullifnull@TransactionalpublicvoidfetchUsers()
varusers=publicvoidfetchSortedProducts()varsort="name""price"publicvoid

```

```

fetchPaginatedProducts(intintPageRequestpageRequest=varproducts=vartotalPages=
vartotalElements="Total Pages: ""Total Elements: "@Transactionalpublicvoid
printLoyalProfiles()varusers=2": " com.codewithmosh.store.services;

```

```

com.codewithmosh.store.entities.*;
com.codewithmosh.store.repositories.AddressRepository;
com.codewithmosh.store.repositories.ProfileRepository;
com.codewithmosh.store.repositories.UserRepository;
com.codewithmosh.store.repositories.specifications.ProductSpec;
jakarta.persistence.EntityManager;
jakarta.transaction.Transactional;
lombok.AllArgsConstructor;
org.springframework.data.domain.*;
org.springframework.data.jpa.domain.Specification;
org.springframework.stereotype.Service;

```

```

java.math.BigDecimal;

```

```

{
 UserRepository userRepository;
 ProfileRepository profileRepository;
 EntityManager entityManager;
 AddressRepository addressRepository;
 ProductRepository productRepository;
 CategoryRepository categoryRepository;

```

```

{
 User.builder()
 .name()
 .email()
 .password()
 .build();

```

```

(entityManager.contains(user))
 System.out.println();

```

```

 System.out.println();

```

```

userRepository.save(user);

```

```

(entityManager.contains(user))
 System.out.println();

```

```

 System.out.println();
 }

 {
 profileRepository.findById().orElseThrow();
 System.out.println(profile.getUser().getEmail());
 }

 {
 addressRepository.findById().orElseThrow();
 }

 {
 User.builder()
 .name()
 .email()
 .password()
 .build();

 Address.builder()
 .street()
 .city()
 .state()
 .zip()
 .build();

 user.addAddress(address);

 userRepository.save(user);
 }

 {
 userRepository.findById().orElseThrow();
 user.getAddresses().getFirst();
 user.removeAddress(address);
 userRepository.save(user);
 }

 {
 productRepository.deleteById();
 }

```

```

{
 productRepository.updatePriceByCategory(BigDecimal.valueOf(), ());
}

{
 ();
 product.setName();

 ExampleMatcher.matching()
 .withIncludeNullValues()
 .withIgnorePaths(,)
 .withStringMatcher(ExampleMatcher.StringMatcher.CONTAINING);

 Example.of(product, matcher);

 productRepository.findAll(example);
 products.forEach(System.out::println);
}

{
 productRepository.findProductsByCriteria(, BigDecimal.valueOf(),);
 products.forEach(System.out::println);
}

{
 Specification<Product> spec = Specification.where();

 (name !=) {
 spec = spec.and(ProductSpec.hasName(name));
 }
 (minPrice !=) {
 spec = spec.and(ProductSpec.hasPriceGreaterThanOrEqualTo(minPrice));
 }
 (maxPrice !=) {
 spec = spec.and(ProductSpec.hasPriceLessThanOrEqualTo(maxPrice));
 }

 productRepository.findAll(spec).forEach(System.out::println);
}

{
 userRepository.findAllWithTags();
}

```

```

 users.forEach(u -> {
 System.out.println(u);
 u.getAddresses().forEach(System.out::println);
 });
 }

 {
 Sort.by().and(
 Sort.by().descending()
);

 productRepository.findAll(sort).forEach(System.out::println);
 }

 pageNumber, size) {
 PageRequest.of(pageNumber, size);
 Page<Product> page = productRepository.findAll(pageRequest);

 page.getContent();
 products.forEach(System.out::println);

 page.getTotalPages();
 page.getTotalElements();
 System.out.println(+ totalPages);
 System.out.println(+ totalElements);
 }

 {
 userRepository.findLoyalUsers();
 users.forEach(p -> System.out.println(p.getId() + + p.getEmail()));
 }
}
package import import import import import @SpringBootApplication public class
StoreApplication public static void main(String[] args) {
 ApplicationContext context = var service
=010 com.codewithmosh.store;

 com.codewithmosh.store.services.UserService;
 org.springframework.boot.SpringApplication;
 org.springframework.boot.autoconfigure.SpringBootApplication;
 org.springframework.context.ApplicationContext;

 java.math.BigDecimal;

 {

```

```

 {
 SpringApplication.run(StoreApplication.class, args);
 context.getBean(UserService.class);
 service.fetchPaginatedProducts(,);
 }
}
spring:application:name:storedatasource:url:jdbc:mysql://localhost:3306/store?
createDatabaseIfNotExist=trueusername:rootpassword:MyPassword!jpa:show-sql:true

```

```

CREATEPROCEDUREDECIMAL102DECIMAL102BEGINSELECTFROMWHERE
BETWEENANDORDERBYENDDELIMITER $$

```

```

findProductsByPrice(
 minPrice (,),
 maxPrice (,)
)

```

```

 id, name, description, price, category_id
products
price minPrice maxPrice
name;

```

\$\$

```

DELIMITER ;create tablebigintprimary keyvarchar255not nullvarchar255not nullvarchar
255not nullcreate tablebigintprimary keyvarchar255not nullvarchar255not nullvarchar
255not nullbigintnot nullconstraintforeign keyreferences users

```

```

(
 id auto_increment
,
 name (),
 email (),
 password ()
);

```

addresses



```
(
 id auto_increment
 ,
 street () ,
 city () ,
 zip () ,
 user_id ,
 addresses_users_id_fk
 (user_id) users (id)
);
```

```
alter table add varchar 255 null users
state ();
```

```
alter table drop column alter table add varchar 255 not null users
state;
```

```
addresses
state ();
```

```
CREATE TABLE BIGINT PRIMARY KEY VARCHAR 15 DATE INT DEFAULT 0 FOREIGN
KEY REFERENCES profiles
```

```
(
 id ,
 bio TEXT,
 phone_number (),
 date_of_birth ,
 loyalty_points UNSIGNED ,
 (id) users(id)
);
```

```
-- Create the tags table CREATE TABLE INT PRIMARY KEY VARCHAR 255 NOT NULL --
Create the user_tags join table CREATE TABLE BIGINT NOT NULL INT NOT NULL
PRIMARY KEY FOREIGN KEY REFERENCES ON DELETE FOREIGN KEY
REFERENCES ON DELETE
```

```
tags
(
 id AUTO_INCREMENT ,
 name ()
);
```

```
user_tags
(
 user_id ,
 tag_id ,
 (user_id, tag_id),
```

```

 (user_id) users (id) CASCADE,
 (tag_id) tags (id) CASCADE
);
CREATE TABLE PRIMARY KEY VARCHAR(255) NOT NULL CREATE TABLE BIGINT
PRIMARY KEY VARCHAR(255) NOT NULL DECIMAL(10,2) NOT NULL CONSTRAINT
FOREIGN KEY REFERENCES ON DELETE categories
(
 id TINYINT AUTO_INCREMENT ,
 name ()
);

products
(
 id AUTO_INCREMENT ,
 name () ,
 price (,) ,
 category_id TINYINT,
 fk_category
 (category_id) categories (id)
 RESTRICT
);
ALTER TABLE ADD NULL ALTER TABLE NOT NULL products
`description` TEXT ;

products
MODIFY `description` TEXT ; CREATE TABLE BIGINT NOT NULL BIGINT NOT NULL
CONSTRAINT PRIMARY KEY ALTER TABLE ADD CONSTRAINT FOREIGN KEY
REFERENCES ALTER TABLE ADD CONSTRAINT FOREIGN KEY REFERENCES
wishlist
(
 product_id ,
 user_id ,
 pk_wishlist (product_id, user_id)
);

wishlist
 fk_wishlist_on_product (product_id) products (id);

wishlist
 fk_wishlist_on_user (user_id) users (id); alter table drop foreign key alter table add
constraint foreign key references on delete wishlist
 fk_wishlist_on_product;

wishlist
 fk_wishlist_on_product
 (product_id) products (id)

```

```
cascade;
```

```
<!DOCTYPE html<html lang="en"<head<meta charset="UTF-8"<title</title</head<body<h1</h1</body</html>>
```

```
=>
```

```
>
```

```
 = />
```

```
 >Title>
```

```
>
```

```
>
```

```
 >Hello World>
```

```
>
```

```
>
```

```
package import import @SpringBootTest class StoreApplicationTests @Test void
```

```
contextLoads() { com.codewithmosh.store;
```

```
 org.junit.jupiter.api.Test;
```

```
 org.springframework.boot.test.context.SpringBootTest;
```

```
{
```

```
 {
```

```
 }
```

```
}
```

